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Action oriented detection of synthetic media objects

Abstract

A synthetic media detection system stores, for each of a set of predefined objects an object repository with previously determined characteristics of predefined objects and an action description repository with natural language descriptions of actions that are known to be possible to occur with the predefined object. The system receives at least a portion of a media stream including a video to be presented on a media player device. An object is detected in the received portion of the media stream and properties of the detected object are determined. An action is determined that is associated with the detected object. An object probability is determined that the detected object is a synthetic media object. A chronological description is determined of the actions associated with the detected object. An overall probability that the object is synthetic media object using the chronological description is determined and displayed with the video.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to synthetic media. More particularly, in certain embodiments, the present disclosure is related to action oriented detection of synthetic media objects.

BACKGROUND

(2) Synthetic media, such as so called “deepfake” videos, can be generated to mislead media consumers. For instance, a video of an individual speaking can be edited or generated to make it appear as though a person has spoken words that they did not actually speak and/or performed actions they did not actually perform.

SUMMARY

(3) Synthetic media is becoming increasingly realistic and difficult to recognize. Humans are generally unable to reliably identify images and videos that contain synthetic media. It is difficult to reliably and efficiently detecting synthetic media and effectively informing users that they are potentially viewing synthetic media. This disclosure recognizes that previous tools fail to provide for the detection of synthetic objects in media, such as any item or other object that may be artificially added to a video or image to create a scenario that did not occur in the real world. For example, previous technology fails to provide efficient and reliable tools for detecting synthetic media in which an object is artificially added or otherwise modified.

(4) Certain embodiments of this disclosure are integrated into the practical application of an action-based synthetic media detection system that provides unique solutions to technical problems of previous technology, including those described above, by efficiently and reliably detecting synthetic objects in media. For example, the disclosed system provides several technical advantages which may include 1) the efficient and reliable detection of synthetic objects in media based at least in part on actions occurring with or around the objects; 2) the automatic implementation of a two-prong probability determination that not only accounts for the appearance of objects in images but also evaluates an automatically generated description of an object's actions to determine a more reliable assessment of synthetic media than was previously possible; and 3) the presentation of a dynamically updated indicator of the probability that displayed media includes a synthetic media object. As such, this disclosure may improve the function of computer systems used to present media, detect synthetic media, and/or report the presence of suspected synthetic media. For example, the system described in this disclosure may decrease processing resources required to review media and improve the reliability of the results of this review (e.g., by adjusting the frequency of media stream sampling based on a current synthetic media probability level in certain embodiments).

(5) Certain embodiments of this disclosure may include some, all, or none of these advantages. These advantages and other features will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings and claims.

(6) In an embodiment, a synthetic media detection system stores, for each of a set of predefined objects, (1) an object repository with previously determined characteristics of predefined objects and (2) an action description repository with natural language descriptions of actions that are known to be possible to occur with the predefined object. The synthetic media detection system receives at least a portion of a media stream that includes a video to be presented on a media player device. An object is detected in the received portion of the media stream. Properties of the detected object are determined. An action is determined that is associated with the detected object. An object probability is determined that the detected object is a synthetic media object that is artificially included in the video based on a comparison of the properties of the detected object to the

previously determined characteristics of predefined objects of the object repository. A chronological description is determined of the actions associated with the detected object. An overall probability that the detected object is the synthetic media object is determined based on whether the chronological description includes actions that can occur with the detected object. The media presentation device displays the overall probability along with the video.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of this disclosure, reference is now made to the following brief description, taken in connection with the accompanying drawings and detailed description, wherein like reference numerals represent like parts.

(2) FIG. **1** is a schematic diagram of an example synthetic media detection system; and

(3) FIG. **2** is a flowchart illustrating an example method of operating the system of FIG. **1**.

DETAILED DESCRIPTION

(4) As described above, prior to this disclosure, there was a lack of tools for reliably detecting synthetic media objects and presenting results of this detection. The system described in this disclosure solves technical problems of previous technology by detecting synthetic media objects using an action-oriented approach. The system also facilitates the presentation of a dynamically determined probability that viewed media includes synthetic media.

(5) As used in this disclosure, media generally refers to video, images, and/or audio content. However, media encompasses any other appropriate media which may be shared and/or distributed in a digital format. An example of media for a subject is a video of the subject speaking. As used in this disclosure, “real media” refers to media that depicts (e.g., is a recording or other representation of) the subject without any adulteration to the information provided in the media. For example, a “real” video of a subject may be an audiovisual recording of the subject speaking. As used in this disclosure, “synthetic media” refers to media which has been edited to attribute actions and/or words to a subject that were not performed/spoken by the subject. For example, a “synthetic” video may include an edited version of a “real” video of the subject speaking which has been created or edited to show the subject speaking words that were not actually spoken by the subject in the real video.

(6) System for Synthetic Media Detection and Reporting

(7) FIG. **1** is a schematic diagram of an example system **100** for detecting potential synthetic media in a media stream **104**. The system **100** includes one or more media stream sources **102**, a media player device **106**, and an action-based detection system **112**. The one or more media stream sources **102** generally include any source(s) of a media stream **104**, such as a server, data store, or database. For instance, the media stream **104** may be provided or retrieved over a network from a server media stream source **102** configured to host the media stream **104**. The media stream **104** may include any type of media such as one or more videos **146** (e.g., audio-visual recordings), audio recordings, images, and the like. For example, the media stream **104** may include an audio-visual recording or video that includes an object **150**.

(8) The media player device **106** is generally any device operable to receive the media stream **104** and display a video **146** included in the media stream **104**. The media player device **106** includes a video-audio sink **108** and a video player engine **144**. The media player device **106** and its various components **108**, **144** may be implemented using processor **152**, memory **154**, and device network interface **156** described further below. The video-audio sink **108** uses memory **154** of the media player device **106** to store, at least temporarily, the media stream **104** as it is received. The video-audio sink **108** provides the media stream **104** to the video player engine **144** and provides at least a portion of the media stream **104** as frame group buffer **110** to the action-based detection system

112. The video-audio sink **108** may temporarily delay provisioning of the media stream **104** to the video player engine **144**, such that operations of the action-based detection system **112** may be performed to determine an overall probability **164** that the media stream **104** includes synthetic media before the video **146** is displayed. The frame group buffer **110** may be stored in memory **154** of the media player device **106** as a portion of image frames of the media stream **104**.
(9) The frame group buffer **110** of the media stream **104** is selected and provided to the action-based detection system **112**, which uses this information to determine an overall probability **164** that the object **150** appearing in video **146** is a synthetic media object (as described in greater detail below). In some embodiments, the amount of information or the frequency of information provided in the frame group buffer **110** is adjusted based on a previous overall probability **164** determined by the action-based detection system **112**. For example, when the previous overall probability **164** that an object **150** is synthetic media is low, then the amount of data sent in the frame group buffer **110** may decreased to a predefined low level (e.g., of a certain number of image frames per unit of time). However, when the overall probability **164** exceeds a threshold **166** (e.g., a threshold value), the amount of data sent in the frame group buffer **110** may be increased to increase the reliability and better refine the evaluations performed by the action-based detection system **112**.
(10) The video player engine **144** of the media player device **106** receives the media stream **104** and the synthetic media overall probability **164** and displays both. For example, the video player engine **144** may display the media stream **104** as displayed video **146** with an overlay of the synthetic media overall probability **164**. Thus, the media player device **106** includes the technical improvement of a dynamically updated indicator of the overall probability **164** that displayed video **146** includes an object **150** that is synthetic media (e.g., that did not originally appear in the video **146** and/or did not behave in the real world as depicted in the video **146**). The object **150** may be highlighted in the video **146** (e.g., by a circle or box around the object or any other appropriate visual cue) to indicate that the overall probability **164** corresponds to the likelihood that the particular object **150** was added/modified as synthetic media to the video **146**. In some embodiments, the synthetic media overall probability **164** is added as a watermark to the displayed video **146**.
(11) In some embodiments, a toggle button **148** or other option field may be presented in or around the displayed video **146** in order for a user to request that the overall probability **164** be presented or hidden. As such, the video media player device **106** may receive a user input based on operation of the toggle button **148**, determine if the user input corresponds to a request to display or hide the overall probability **164**, and either display or hide the overall probability **164** based on this request. In some cases, the overall probability **164** is only determined by the action-based detection system **112** if a user requests display of the overall probability **164**, thereby conserving processing resources of the action-based detection system **112**. In other cases, the overall probability **164** is determined even when display of the overall probability **164** is not currently requested (e.g., but optionally at a lower rate of acquisition of the frame group buffer **110** to partially decrease computational costs). This allows the overall probability **164** to be immediately available for display when such display is requested via toggle button **148**. When display is requested, the frame group buffer **110** may be provided at a higher frequency for improved accuracy of the overall probability **164** over time.
(12) As described above, the media player device **106** includes a processor **152**, memory **154**, and the device network interface **156**. The processor **152** includes one or more processors. The processor **152** is any electronic circuitry including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g., a multi-core processor), field-programmable gate array (FPGAs), application specific integrated circuits (ASICs), or digital signal processors (DSPs). The processor **152** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The processor **152** is communicatively coupled to and in signal communication with the memory **154** and device

network interface **156**. The one or more processors are configured to process data and may be implemented in hardware and/or software. For example, the processor **152** may be 8-bit, 16-bit, 32-bit, 64-bit or of any other suitable architecture. The processor **152** may include an arithmetic logic unit (ALU) for performing arithmetic and logic operations, processor registers that supply operands to the ALU and store the results of ALU operations, and a control unit that fetches instructions from memory **154** and executes them by directing the coordinated operations of the ALU, registers and other components.

(13) The memory **154** is operable to store any data, instructions, logic, rules, or code operable to execute the functions of the media player device **106**. The memory **154** includes one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. The memory **154** may be volatile or non-volatile and may comprise read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM).

(14) The device network interface **156** is configured to enable wired and/or wireless communications. The device network interface **156** is configured to communicate data between the media player device **106** and other network devices, systems, or domain(s), such as the media stream source **102** and action-based detection system **112**. The device network interface **156** is an electronic circuit that is configured to enable communications between devices. For example, the device network interface **156** may include one or more serial ports (e.g., USB ports or the like) and/or parallel ports (e.g., any type of multi-pin port) for facilitating this communication. As a further example, the device network interface **156** may include a WIFI interface, a local area network (LAN) interface, a wide area network (WAN) interface, a modem, a switch, or a router. The processor **152** is configured to send and receive data using the device network interface **156**. The device network interface **156** may be configured to use any suitable type of communication protocol.

(15) The action-based detection system **112** is generally any device or collection of devices operable to determine the overall probability **164** that the object **150** appearing in video **146** is synthetic media. The action-based detection system **112** may be a separate device or collection of devices from the media player device **106** or may be partially or entirely integrated within the media player device **106**. For example, all or a portion of the functions of the action-based detection system **112** may be implemented using an external device or server with the separate processor **158**, memory **160**, and the system network interface **162** described below. In some cases, one or more of the functions of the action-based detection system **112** may be performed by the media player device **106** (e.g., by the processor **152**, memory **154**, and the device network interface **156** of the media player device **106**, described above). Accordingly, in various embodiments, operations described as being performed by the action-based detection system **112** may be performed by the media player device **106** (and vice versa). In some cases, functions requiring more complex computations (e.g., for machine learning algorithms or the like) may be implemented by the separate processor **158**, memory **160**, and system network interface **162** of the action-based detection system **112**, while one or more less computationally costly functions may be executed locally by the media player device **106**.

(16) The action-based detection system **112** includes an object analysis component **114**, an object verification component **126**, and a chronological analyzer **136**, each of which may be implemented by the processor **158** executing instructions stored in memory **160**. The object analysis component **114** generally identifies objects **118***a* (e.g., one or more of the objects **150**) and object properties **118***b* of the object(s) **118***a* appearing in the image frames provided in the frame group buffer **110** along with actions **122** being performed with/by the objects **118***a* in the video **146**. For example, functions of the object analysis component **114** may include tasks illustrated with respect to the object properties identifier **116**, action properties combiner **120** and object action identifier **124** of

FIG. **1**. The object properties identifier **116** generally determines object properties **118***b* of any objects **118***a* detected in the video **146** (e.g., in the portion of the video **146** that is sent to the action-based detection system **112** as frame group buffer **110**). The object properties **118***b* may correspond to a name of a corresponding real-world object **118***a* that is detected in the frame group buffer **110**, properties of the object **118***a* (e.g., size, shape, color, texture, etc.), a class or category of the object **118***a* (e.g., an object **118***a* that is a bottle may be a liquid-holding class of object and an object **118***a* that is a car may be an automobile class of object, etc.), a logo or text appearing on the object **118***a*, and/or any other information that may be used to identify or characterize the object **118***a* detected in the video **146**. The object properties identifier **116** may employ any appropriate object detection and/or characterization algorithms to detect objects **118***a* and determine the object properties **118***b*. Examples of algorithms used for object detection/identification include neural network-based algorithms and the like. For instance, a neural network-based algorithm may employ a convolutional neural network (CNN) that is trained to detect objects **118***a* and determine object properties **118***b* based on properties of previously collected images of the objects **118***a* or similar objects.
(17) The action properties combiner **120** identifies actions **122** appearing in the portion of the video **146** provided in the frame group buffer **110**. Examples of actions include moving, rolling, falling, breaking, driving, floating, hovering, or any other action that may be associated with a given object **118***a*. The action properties combiner **120** may employ any appropriate action detection and/or characterization algorithms to determine the actions **122**. Examples of algorithms used for action detection/identification include neural network-based algorithms, such as CNNs, similarly to as described above for object detection.
(18) The object action identifier **124** determines which of the actions **122** are associated with the detected objects **118***a* and links the actions **122** to the corresponding detected objects **118***a*. The objects **118***a* and their properties **118***b* are used by the object verification component **126** to determine an initial object probability **132** that one or more objects **118***a* may be synthetic media of a known object based on the appearance of the detected objects **118***a*. The chronological analyzer **136** is then used to determine a refined (e.g., overall) probability **164** that one or more objects **118***a* (e.g., object **150** appearing in video **146** in the media player device **106**) may be synthetic media, based at least in part on the actions **122** linked to the objects **118***a*.
(19) The object verification component **126** includes an object originality verifier **128** and an object probability calculator **130**. The object verification component **126** is in communication with an object repository **134**, which stores previously determined characteristics of predefined objects (e.g., sizes/appearances of known objects, texts/logos appearing on known objects, and the like). The object repository **134** may be stored in the memory **160** of the action-based detection system **112**. The object originality verifier **128** determines whether detected objects **118***a* match, or are similar to, known objects in the object repository **134**. The object probability calculator **130** then determines the object probability **132** based on the predefined information in the object repository **134**. The initial object probability **132** provides an initial estimate of whether an object **118***a* is real or synthetic in the media stream **104**, based for example, on a comparison of the object properties **118***b* for a given object **118***a* to the information for the most similar object in the object repository **134**. The object probability **132** may act as a starting point for determining the overall probability **164** by the chronological analyzer **136**, as described below.
(20) This disclosure recognizes that relying on information in the object repository **134** alone may result in certain synthetic media going undetected. To reduce the chance that synthetic media is not detected, the chronological analyzer **136** provides an improved determination of an overall probability **164** that accounts for how detected objects **118***a* behave over time in the video **146**. The chronological analyzer **136** may detect synthetic media that is outside the relatively narrow range of predefined information that can be captured in the object repository **134** by determining whether the chronological series of actions **122** occurring for a given object **118***a* are possible/realistic in the

real world. This possibility is determined by comparing an automatically generated chronological, or frame-wise, description **140** of the object **118***a* and its actions **122** to reference descriptions for that object **118***a* (or a corresponding class of objects) stored in an action description repository **168**.
(21) The chronological analyzer **136** first determines a description **138** of the objects **118***a* and associated actions **122** as text for each frame (or set of frames) corresponding to a given point in time in the video **146**. A machine learning algorithm may be trained to generate the description **138**. The description **138** may be in a natural language. As used in this disclosure, a natural language corresponds to an established language (e.g., English) used for human-to-human communication. Descriptions **138** for different times, or frames, are then transformed through appropriate combination into the chronological description **140** of the object's actions **122** over time in the video **146**. This chronological description **140** is a narrative of the actions **122** occurring with each detected object **118***a* throughout the portion of the video **146** that is provided in the frame group buffer **110** for analysis. As such, in some cases, the amount of data provided in the frame group buffer **110** may be adjusted to ensure the narrative of the chronological description **140** is complete enough for reliable analysis. For example, when the object probability **132** is low or a previously determined overall probability **164** is low (e.g., less than a threshold **166**), less data may be provided in the frame group buffer **110**. However, when there is a need to more accurately determine or verify the presence of a synthetic media object **150** in a video **146**, the data included in the frame group buffer **110** may be increased (e.g., when a high value object **118***a* is detected in a video **146** or when a probability (object probability **132**, overall probability **164**) is relatively high).
(22) The chronological description **140** is then provided to the natural language processing (NLP) evaluator **142**, which compares the chronological description **140** to reference descriptions stored in the action description repository **168** (e.g., which may be stored in memory **160**). The action description repository **168** stores natural language descriptions of actions that are known to be possible to occur for the predefined object. These references may indicate actions that are possible and impossible for certain objects **118***a* or classes of objects **118***a*. For example, a smartphone class object **118***a* is capable of falling if dropped but cannot move on its own. This type of information is captured by the action description repository **168** and used for comparison to the chronological description **140**.
(23) Based on this comparison, an overall probability **164** is then determined. For example, the comparison may indicate whether the chronological description **140** includes actions **122** for an object **118***a* that can occur for the object **118***a*. For example, certain objects may not be capable of independent movement, while other objects are capable of certain types of independent movement. These characteristics are captured by the references in the action description repository **168**. If an object **118***a* is indicated in the chronological description **140** to be making a movement or other action type that is not appropriate for that object **118***a*, then the overall probability **164** that the object **118***a* is synthetic media is increased. Any appropriate method of natural language processing and/or an appropriately trained machine learning algorithm may be used to determine the overall probability **164** using information in the action description repository **168**.
(24) As illustrated in FIG. **1**, the action-based detection system **112** may include a processor **158**, memory **160**, and system network interface **162**. The processor **158** includes one or more processors. The processor **158** is any electronic circuitry including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g., a multi-core processor), field-programmable gate array (FPGAs), application specific integrated circuits (ASICs), or digital signal processors (DSPs). The processor **158** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The processor **158** is communicatively coupled to and in signal communication with the memory **160** and system network interface **162**. The one or more processors are configured to process data and may be implemented in hardware and/or software. For example, the processor **158** may be 8-bit, 16-bit, 32-bit, 64-bit or of any other suitable architecture. The processor **158** may include an

arithmetic logic unit (ALU) for performing arithmetic and logic operations, processor registers that supply operands to the ALU and store the results of ALU operations, and a control unit that fetches instructions from memory **160** and executes them by directing the coordinated operations of the ALU, registers and other components.

(25) The memory **160** is operable to store any data, instructions, logic, rules, or code operable to execute the functions of the action-based detection system **112**. The memory **160** includes one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. The memory **160** may be volatile or non-volatile and may comprise read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM).

(26) The system network interface **162** is configured to enable wired and/or wireless communications. The system network interface **162** is configured to communicate data between the action-based detection system **112** and other network devices, systems, or domain(s), such as the media player device **106**. The system network interface **162** is an electronic circuit that is configured to enable communications between devices. For example, the system network interface **162** may include one or more serial ports (e.g., USB ports or the like) and/or parallel ports (e.g., any type of multi-pin port) for facilitating this communication. As a further example, the system network interface **162** may include a WIFI interface, a local area network (LAN) interface, a wide area network (WAN) interface, a modem, a switch, or a router. The processor **158** is configured to send and receive data using the system network interface **162**. The system network interface **162** may be configured to use any suitable type of communication protocol.

(27) In an example operation of the system **100**, a media stream **104** is received by the media player device **106**. The media stream **104** includes a video **146** (i.e., an audio-visual recording) that includes an artificially added object **150** (i.e., a synthetic media object **150**). In this example, the artificially added object **150** is a smartphone device. The video **146** shows an individual dropping the device from a modest height and the screen of the device shattering. The video-audio sink **108** of the media player device **106** temporarily stores the media stream **104** as it is received and passes a portion of the media stream **104** to the action-based detection system **112** as frame group buffer **110**. Initially, the frame group buffer **110** is provide at a predefined data frequency that is subsequently updated based on determined values of the overall probability **164**.

(28) The action-based detection system **112** identifies a set of objects **118***a* appearing in the video **146** and actions **122** associated with the objects **118***a*. In this example, one detected object **118***a* is the smartphone device. Actions **122** that are determined for the object **118***a* are falling from the height, briefly floating above the ground, and breaking soon after hitting the ground. Object properties **118***b* are used to determine an object probability **132** that is relatively low, indicating that the object **118***a* appears, based on the object properties **118***b*, to be a real smartphone device. However, the chronological analyzer **136** determines a relatively high overall probability **164** that the smartphone device object **118***a* is synthetic media because the brief floating action **122** is not possible in the real world for the smartphone class of object.

(29) The overall probability **164** is presented along with the video **146**. The object **150** in the video **146** may also be flagged (e.g., by a bounding box or other visual indicator) to indicate that the object **150** is associated with the relatively high overall probability **164** of synthetic media. As such, in this example operation, a user viewing the video **146** is informed that the smartphone device depicted as object **150** likely did not really break from the fall despite the video **146** providing a realistic and otherwise seemingly believable presentation of this scenario.

(30) Example Method of Detecting Synthetic Media Objects

(31) FIG. **2** is a flowchart of an example method **200** for operating the system **100** of FIG. **1**. The method **200** generally facilitates the determination of an overall probability **164** that a media stream

104 or video **146** includes synthetic media of an object **150** and the automatic display of this overall probability **164**. The method **200** may be performed by the processor **158**, memory **160**, and the system network interface **162** of the action-based detection system **112** and/or the processor **152**, memory **154**, and the device network interface **156** of the media player device **106**. The method **200** may begin at step **202** where one or more objects **118***a* are detected in the video **146** (e.g., in the portion of the video **146** provided in the frame group buffer **110**). The action-based detection system **112** may receive at least a portion of a media stream comprising a video to be presented on the media player device **106** and detect object(s) **118***a* in the portion received, for example, as described with respect to the object analysis component **114** of FIG. **1**. At step **204**, the action-based detection system **112** determines whether at least one object **118***a* is detected. If at least one object **118***a* is detected, the action-based detection system **112** proceeds to step **206**.

(32) At step **206**, object properties **118***b* are determined for each detected object **118***a*. The object properties **118***b* generally describe features of the detected objects **118***a*, such as size (e.g., length, width, height, apparent volume, etc.) of the detected objects **118***a*, color of the detected objects **118***a*, a logo appearing on the detected objects **118***a*, text appearing on the detected object **118***a*, an object class of the detected objects **118***a*, and the like. The object properties **118***b* may be determined as described above with respect to the object analysis component **114** of FIG. **1**.

(33) At step **208**, the action-based detection system **112** determines one or more actions **122** associated with each detected object **118***a*. Examples of actions **122** include moving, rolling, falling, breaking, driving, floating, hovering, or any other action that may be associated with a given object **118***a*. Any appropriate action detection and/or characterization algorithm(s) may be used to determine the object properties **118**, as described above with respect to the example of the object analysis component **114** of FIG. **1**.

(34) At step **210**, an object probability **132** is determined for each detected object **118***a*. The object probability **132** corresponds to an initial percentage likelihood that the object **118***a* is a synthetic media object that has been artificially included in the video **146**. The object probability **132** may be determined based on a comparison of the object properties **118***b* of the detected object **118***a* to previously determined characteristics of predefined objects stored in the object repository **134**, as described in greater detail above with respect to FIG. **1**. In some cases, if the object probability **132** is high (e.g., greater than a threshold **166**), the action-based detection system may determine that the object **118***a* is very likely a synthetic media object and chronological analysis (e.g., of steps **212-216** may not need to be performed). However, in typical embodiments, chronological analysis is still performed to help ensure that well-designed synthetic media objects that have properties **118***b* similar to those of their real-world counterparts do not go undetected.

(35) At step **212**, a chronological natural language description **140** is determined for the video **146** (e.g., or the portion of the video **146** that has been received for analysis via frame group buffer **110**). The chronological description **140** includes a narrative of the actions **122** occurring with each detected object **118***a* throughout the portion of the video **146** that is provided in the frame group buffer **110** for analysis. At step **214**, the chronological description **140** is compared to one or more reference descriptions in the action description repository **168**. The action description repository **168** stores natural language descriptions of actions that are known to be possible to occur with real-world objects. These references may indicate actions that are possible and/or impossible for certain objects **118***a* or classes of objects **118***a* in the real world. For example, a smartphone class object **118***a* is capable of falling if dropped but cannot float or glide to a soft landing. This type of information is captured by the action description repository **168** and used for comparison to the chronological description **140**.

(36) At step **216**, the overall probability **164** that an object **118***a* is a synthetic media object is determined based on the comparison at step **214**. For example, the comparison from step **214** may indicate whether the chronological description **140** includes actions **122** for an object **118***a* that can occur in the real world with the object **118***a*. For example, certain objects may not be capable of

independent movement, while other objects are capable of certain types of independent movement. These characteristics are captured by the references in the action description repository **168**. If an object **118***a* is indicated in the chronological description **140** to be involved with an action **122** that is not appropriate or possible for that object **118***a* in the real world, then the overall probability **164** that the object **118***a* is synthetic media is increased. Any appropriate method of natural language processing and/or an appropriately trained machine learning algorithm may be used to determine the overall probability **164**.

(37) At step **218**, the media presentation device **106** displays the overall probability **164** with the video **146**, as described above with respect to FIG. **1**. In some cases, a selectable toggle button **148** can be adjusted via a user input to prevent or allow display of the overall probability **164**. In some cases, the selection at toggle button **148** may impact how/whether the overall probability **164** is determined, providing further improvements in the efficiency with which computational resources are used to determine the overall probability **164** and/or the speed at which the overall probability **164** can be determined. For example, when the toggle button **148** is adjusted to prevent display of the overall probability **164**, the action-based detection system **112** may at least temporarily stop determining the overall probability **164**, such that computing resources are not expended when display of the overall probability **164** is not requested. However, in other cases (e.g., when it is desirable to have near instant access to the overall probability **164** when it is requested), when the toggle button **148** is adjusted to prevent display of the overall probability **164**, the action-based detection system **112** continues determining the overall probability **164**, such that the overall probability **164** is immediately or nearly immediately available when the toggle button **148** is subsequently selected to allow presentation of the overall probability **164**.

(38) At step **220**, the action-based detection system **112** determines whether overall probability **164** is less than a threshold **166**. If this is the case, the action-based detection system **112** may decrease the rate or frequency at which data is provided in the frame group buffer **110** at step **222**. In other words, the action-based detection system **112** may select the portion of the media stream **104** to receive in frame group buffer **110** based at least in part on the overall probability **164**. When the overall probability **164** is relatively low (e.g., less than threshold **166**), a reduced portion of the media stream **104** is received to reduce consumption of processing resources by the action-based detection system when there is a low initial likelihood of synthetic media. If the overall probability **164** increases (e.g., as more frames of the video **146** are analyzed), the data provided in the frame group buffer **110** can be increased to improve synthetic media detection in subsequent portions (e.g., frames or time points) of the video **146**.

(39) While several embodiments have been provided in this disclosure, it should be understood that the disclosed systems and methods might be embodied in many other specific forms without departing from the spirit or scope of this disclosure. The present examples are to be considered as illustrative and not restrictive, and the intention is not to be limited to the details given herein. For example, the various elements or components may be combined or integrated in another system or certain features may be omitted, or not implemented.

(40) In addition, techniques, systems, subsystems, and methods described and illustrated in the various embodiments as discrete or separate may be combined or integrated with other systems, modules, techniques, or methods without departing from the scope of this disclosure. Other items shown or discussed as coupled or directly coupled or communicating with each other may be indirectly coupled or communicating through some interface, device, or intermediate component whether electrically, mechanically, or otherwise. Other examples of changes, substitutions, and alterations are ascertainable by one skilled in the art and could be made without departing from the spirit and scope disclosed herein.

(41) To aid the Patent Office, and any readers of any patent issued on this application in interpreting the claims appended hereto, applicants note that they do not intend any of the appended

claims to invoke 35 U.S.C. § 112(f) as it exists on the date of filing hereof unless the words “means for” or “step for” are explicitly used in the particular claim.

Claims

1. A system comprising: a memory configured to store: an object repository comprising a plurality of previously determined characteristics of an object; and a plurality of descriptions of possible actions associated with the object; a processor communicatively coupled to the memory and configured to: receive a media stream comprising a plurality of portions of a video to be presented on a media player device; electronically extract a first portion of the plurality of portions from the media stream; include, based at least in part upon a previous probability for synthetic objects in the media stream, the first portion of the media stream in a frame group buffer, wherein: the frame group buffer comprises more data when the previous probability for synthetic objects in the media stream is above a threshold; and the frame group buffer comprises less data when the previous probability for the synthetic objects in the media stream is below the threshold; detect the object in the first portion of the media stream; electronically retrieve a plurality of object properties associated with the object in the first portion of the media stream; identify, appearing in the first portion of the media stream, an action associated with the object; compare the plurality of object properties to the plurality of previously determined characteristics of the object; assign a probability that the object in the first portion of the media stream is artificially included in the video to resemble the object based on a comparison of the plurality of object properties to the plurality of previously determined characteristics of the object; generate a chronological description of a plurality of actions associated with the object in the video; electronically verify whether the chronological description of the plurality of actions associated with the object in the video matches the plurality of descriptions of the possible actions associated with the object; in response to electronically verifying that the chronological description of the plurality of actions associated with the object in the video do not match the plurality of descriptions of the possible actions associated with the object, delay provisioning of the media stream to the media player device; determine that the chronological description of the plurality of actions associated with the object in the video includes actions that cannot occur in association with the object in the video; in response to determining that the chronological description of the plurality of actions associated with the object in the video includes actions that cannot occur in association with the object in the video, assign a refined probability that the object is artificially included in the video to resemble the object based at least in part upon a number of actions that cannot occur in association with the object in the video; verify whether the refined probability is greater than a threshold, the refined probability being an updated version of the probability; in response to electronically verifying that the refined probability is greater than the threshold, flag that the object within the first portion of the media stream is synthetic, a flagged version of the object in the first portion of the media stream comprising a visual indicator that the object is synthetic; electronically interpolate the flagged version of the object into the first portion of the media stream; and transmit, to the media player device, the first portion of the media stream in the frame group buffer, the first portion comprising the flagged version of the object; and the media player device comprising a screen configured to display the flagged version of the object in the first portion of the media stream along the refined probability, wherein the media player device displays the first portion of the media stream preceded by the delay.
2. The system of claim 1, wherein the processor is further configured to: select the first portion of the media stream to receive based at least in part upon the refined probability, wherein when the refined probability is less than a threshold value, a reduced portion of the media stream is received to reduce consumption of processing resources.
3. The system of claim 1, wherein the plurality of object properties comprise one or more of a size

of the object, a color of the object, a logo appearing on the object, text appearing on the object, and an object class of the object.

4. The system of claim 1, wherein the media player device is further configured to: provide a selectable toggle button configured to block or allow display of the refined probability.

5. The system of claim 4, wherein the processor is further configured to: when the selectable toggle button is adjusted to block display of the refined probability, at least temporarily stop determining the refined probability, such that computing resources are not expended when display of the refined probability is not requested.

6. The system of claim 4, wherein the processor is further configured to: when the selectable toggle button is adjusted to block display of the refined probability, continue determining the refined probability, such that the refined probability is immediately available when the selectable toggle button is subsequently selected to allow presentation of the refined probability.

7. The system of claim 1, wherein the processor is further configured to determine the chronological description of the actions associated with the object using a method of natural language processing.

8. A method comprising: receiving a media stream comprising a plurality of portions of a video to be presented on a media player device; electronically extracting a first portion of the plurality of portions from the media stream; including, based at least in part upon a previous probability for synthetic objects in the media stream, the first portion of the media stream in a frame group buffer, wherein: the frame group buffer comprises more data when the previous probability for synthetic objects in the media stream is above a threshold; and the frame group buffer comprises less data when the previous probability for the synthetic objects in the media stream is below the threshold; detecting an object in the first portion of the media stream; electronically retrieving a plurality of object properties associated with the object in the first portion of the media stream; identifying, appearing in the first portion of the media stream, an action associated with the object; comparing the plurality of object properties to a plurality of previously determined characteristics of the object; assigning a probability that the object in the first portion of the media stream is artificially included in the video to resemble the object based on a comparison of the plurality of object properties to the plurality of previously determined characteristics of the object; generating a chronological description of a plurality of actions associated with the object in the video; electronically verifying whether the chronological description of the plurality of actions associated with the object in the video matches a plurality of descriptions of possible actions associated with the object; in response to electronically verifying that the chronological description of the plurality of actions associated with the object in the video do not match the plurality of descriptions of the possible actions associated with the object, delay provisioning of the media stream to the media player device; determining that the chronological description of the plurality of actions associated with the object in the video includes actions that cannot occur in association with the object in the video; in response to determining that the chronological description of the plurality of actions associated with the object in the video includes actions that cannot occur in association with the object in the video, assigning a refined probability that the object is artificially included in the video to resemble the object based at least in part upon a number of actions that cannot occur in association with the object in the video; verifying whether the refined probability is greater than a threshold, the refined probability being an updated version of the probability; in response to electronically verifying that the refined probability is greater than the threshold, flagging that the object within the first portion of the media stream is synthetic, a flagged version of the object in the first portion of the media stream comprising a visual indicator that the object is synthetic; electronically interpolating the flagged version of the object into the first portion of the media stream; transmitting, to the media player device, the first portion of the media stream in the frame group buffer, the first portion comprising the flagged version of the object; and displaying, using the media player device comprising a screen, the flagged version of the object in the first portion of

the media stream along the refined probability, wherein the media player device displays the first portion of the media stream preceded by the delay.
9. The method of claim 8, further comprising: selecting the first portion of the media stream to receive based at least in part upon the refined probability, wherein when the refined probability is less than a threshold value, a reduced portion of the media stream is received to reduce consumption of processing resources.
10. The method of claim 8, wherein the plurality of object properties comprise one or more of a size of the object, a color of the object, a logo appearing on the object, text appearing on the object, and an object class of the object.
11. The method of claim 8, further comprising providing a selectable toggle button configured to block or allow displaying of the refined probability.
12. The method of claim 11, further comprising: when the selectable toggle button is adjusted to block displaying of the refined probability, at least temporarily stopping determination of the refined probability, such that computing resources are not expended when displaying of the refined probability is not requested.
13. The method of claim 11, wherein, when the selectable toggle button is adjusted to block display of the refined probability, continuing to determine the refined probability, such that the refined probability is immediately available when the selectable toggle button is subsequently selected to allow presentation of the refined probability.
14. The method of claim 8, further comprising: determining the chronological description of the actions associated with the object using a natural language processing process.
15. A non-transitory computer-readable medium storing instructions that when executed by a processor cause the processor to: receive a media stream comprising a plurality of portions of a video to be presented on a media player device; electronically extract a first portion of the plurality of portions from the media stream; include, based at least in part upon a previous probability for synthetic objects in the media stream, the first portion of the media stream in a frame group buffer, wherein: the frame group buffer comprises more data when the previous probability for synthetic objects in the media stream is above a threshold; and the frame group buffer comprises less data when the previous probability for the synthetic objects in the media stream is below the threshold; detect an object in the first portion of the media stream; electronically retrieve a plurality of object properties associated with the object in the first portion of the media stream; identify, appearing in the first portion of the media stream, an action associated with the object; compare the plurality of object properties to a plurality of previously determined characteristics of the object; assign a probability that the object in the first portion of the media stream is artificially included in the video to resemble the object based on a comparison of the plurality of object properties of the plurality of object properties to the plurality of previously determined characteristics of the object; generate a chronological description of a plurality of actions associated with the object in the video; electronically verify whether the chronological description of the plurality of actions associated with the object in the video matches a plurality of descriptions of possible actions associated with the object; in response to electronically verifying that the chronological description of the plurality of actions associated with the object in the video do not match the plurality of descriptions of the possible actions associated with the object, delay provisioning of the media stream to the media player device; determine that the chronological description of the plurality of actions associated with the object in the video includes actions that cannot occur in association with the object in the video; in response to determining that the chronological description of the plurality of actions associated with the object in the video includes actions that cannot occur in association with the object in the video, assign a refined probability that the object is artificially included in the video to resemble the object based at least in part upon a number of actions that cannot occur in association with the object in the video; verify whether the refined probability is greater than a threshold, the refined probability being an updated version of the probability; in response to electronically

verifying that the refined probability is greater than the threshold, flag that the object within the first portion of the media stream is synthetic, a flagged version of the object in the first portion of the media stream comprising a visual indicator that the object is synthetic; electronically interpolate the flagged version of the object into the first portion of the media stream; transmit, to the media player device, the first portion of the media stream in the frame group buffer, the first portion comprising the flagged version of the object; and display, using the media player device comprising a screen, the flagged version of the object in the first portion of the media stream along the refined probability, wherein the media player device displays the first portion of the media stream preceded by the delay.

16. The non-transitory computer-readable medium of claim 15, wherein, when executed by the processor, the instructions further cause the processor to: select the first portion of the media stream to receive based at least in part on the refined probability, wherein when the refined probability is less than a threshold value, a reduced portion of the media stream is received to reduce consumption of processing resources.

17. The non-transitory computer-readable medium of claim 15, wherein the plurality of object properties of the object comprises one or more of a size of the object, a color of the object, a logo appearing on the object, text appearing on the object, and an object class of the object.

18. The non-transitory computer-readable medium of claim 15, wherein, when executed by the processor, the instructions further cause the processor to: provide a selectable toggle button configured to block or allow display of the refined probability.

19. The non-transitory computer-readable medium of claim 18, wherein, when executed by the processor, the instructions further cause the processor to: when the selectable toggle button is adjusted to block display of the refined probability, at least temporarily stop determining the refined probability, such that computing resources are not expended when display of the refined probability is not requested.

20. The non-transitory computer-readable medium of claim 18, wherein, when executed by the processor, the instructions further cause the processor to: when the selectable toggle button is adjusted to block display of the refined probability, continue determining the refined probability, such that the refined probability is immediately available when the selectable toggle button is subsequently selected to allow presentation of the refined probability.